Comp9417 Week7

Kernel Methods

Bingchen Xie

Disclaimer:

These slides are my own supplementary resource and are not official course materials. Any mistakes are my own.

Please refer to the solutions posted on **Moodle** as the official source.

Question 1

Question 1 (Dual Perceptron)

Review the development of the Perceptron training algorithm from lectures. Now compare this to the algorithm for Perceptron training *in dual form* introduced in the "Kernel Methods" lecture. The two algorithms are very similar but differ in a few crucial places. Provide an explanation of how the dual version of the algorithm relates to the original.

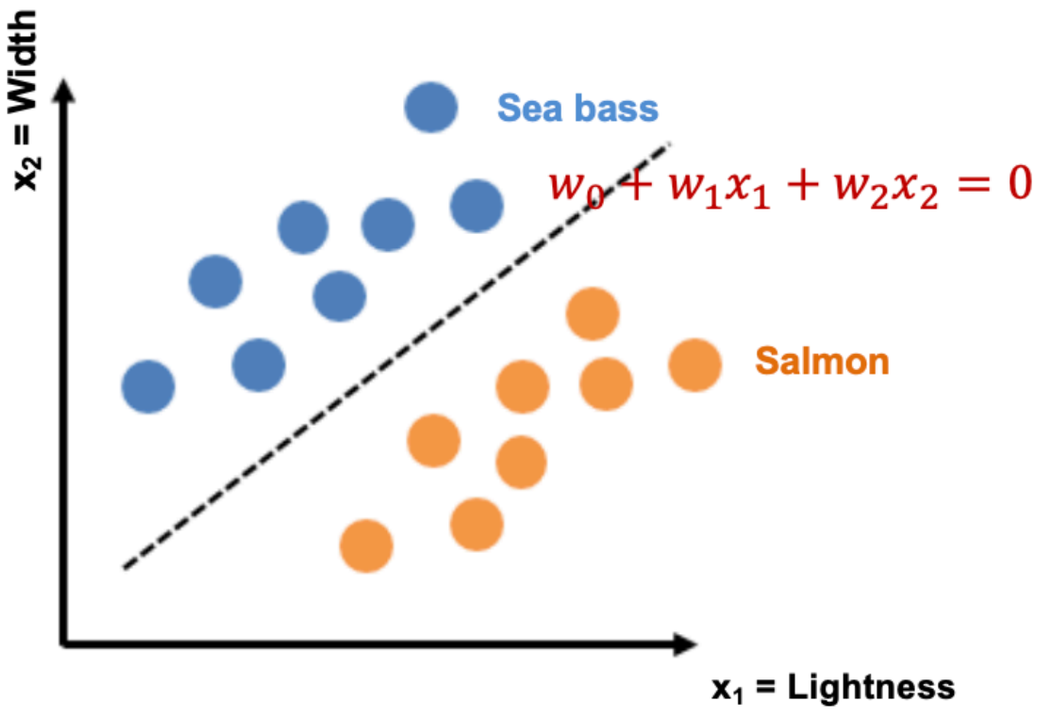
Recap: Linear Classification

Linear classification: Perceptron

Perceptron: is an algorithm for binary classification that uses a linear prediction function

If we have two attributes/features of x_1 and x_2 then we can predict the target function $f(x)$ with:

$$f(\mathrm{x}) = \begin{cases} +1 & if \;\; w_0 + w_1 x_1 + w_2 x_2 > 0 \\ -1 & otherwise. \end{cases}$$



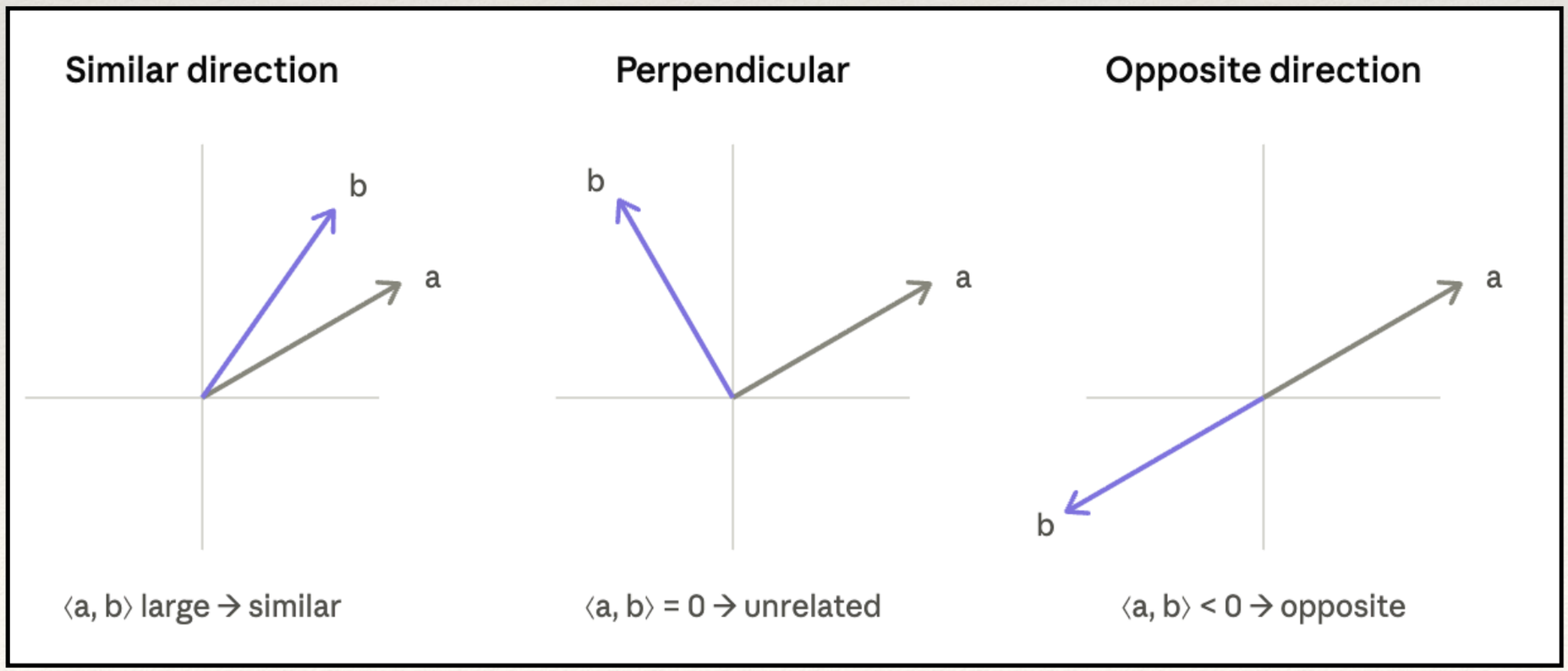
Recap: Inner product

Inner Product (Dot Product): $\langle x, y \rangle = \|x\| \|y\| \cos\theta$

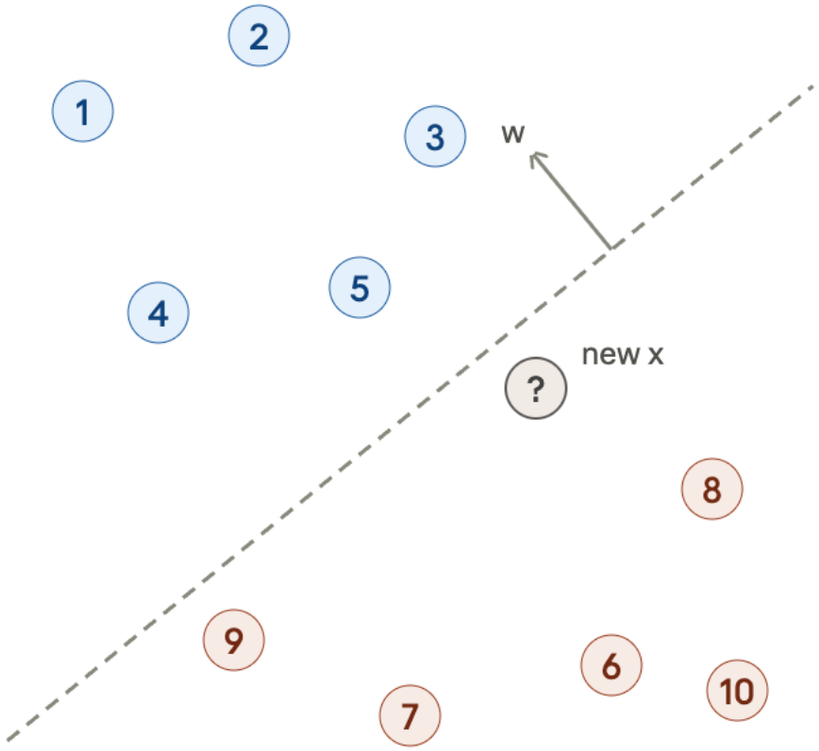
$$\langle a, b \rangle = a_1 b_1 + a_2 b_2 \quad \text{(2D)}$$

$$\langle a, b \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3 \quad \text{(3D)}$$

$$\langle a, b \rangle = a_1 b_1 + \cdots + a_n b_n = \sum_{i=1}^{n} a_i b_i \quad (n\text{-D})$$



Q1 Analyze: Orignal View & Dual View



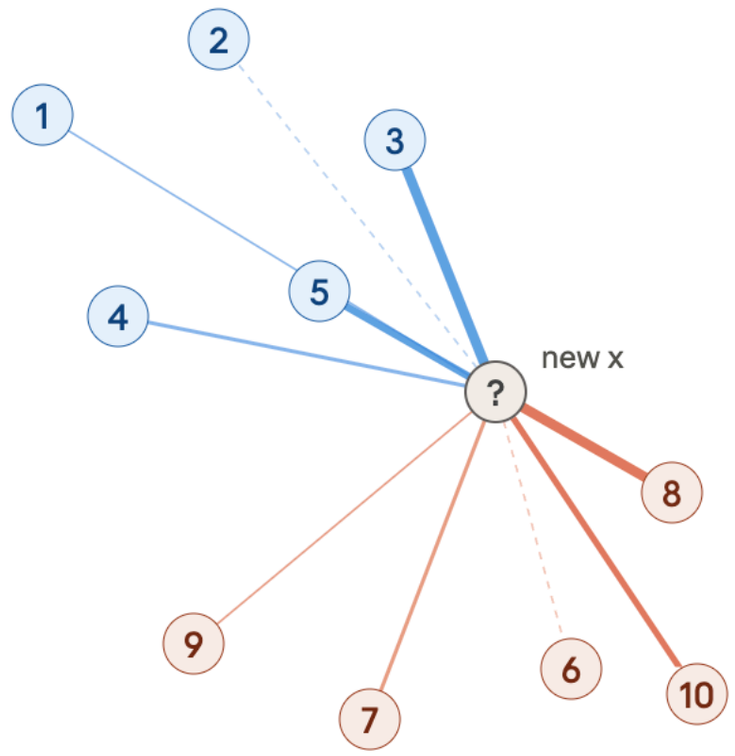
One line, one w
learned from training

↓

Plug in new x
$s = w_1x_1 + w_2x_2 + w_0$

↓

Which side?
$s > 0 \rightarrow +1, s < 0 \rightarrow -1$



Each point's vote
weight α_i × similarity $\langle x_i, x \rangle$

↓

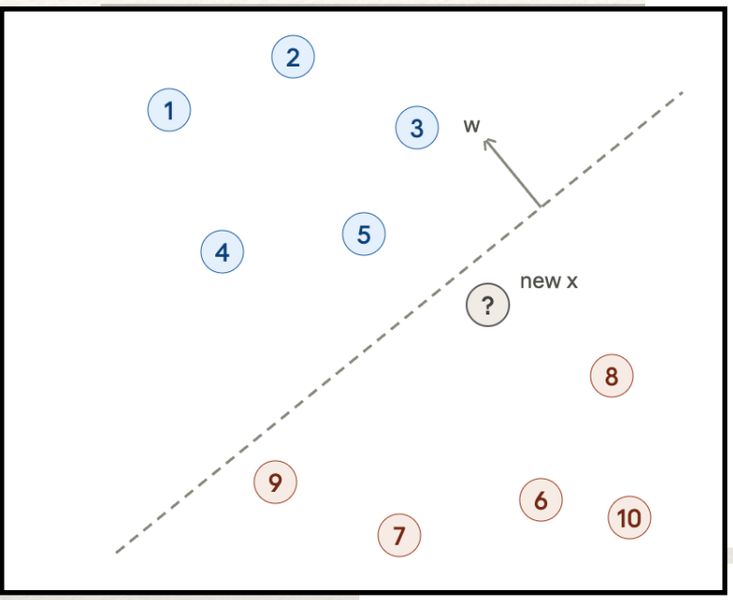
Blue votes
add up points 1 to 5

Orange votes
add up points 6 to 10

↓

Bigger side wins
more blue → +1, more orange → −1

Q1 Analyze: Original View



Perceptron learning

w gets updated only if the prediction mismatches the actual class label (misclassification) and otherwise remains the same. Therefore, for misclassified instances we can write:

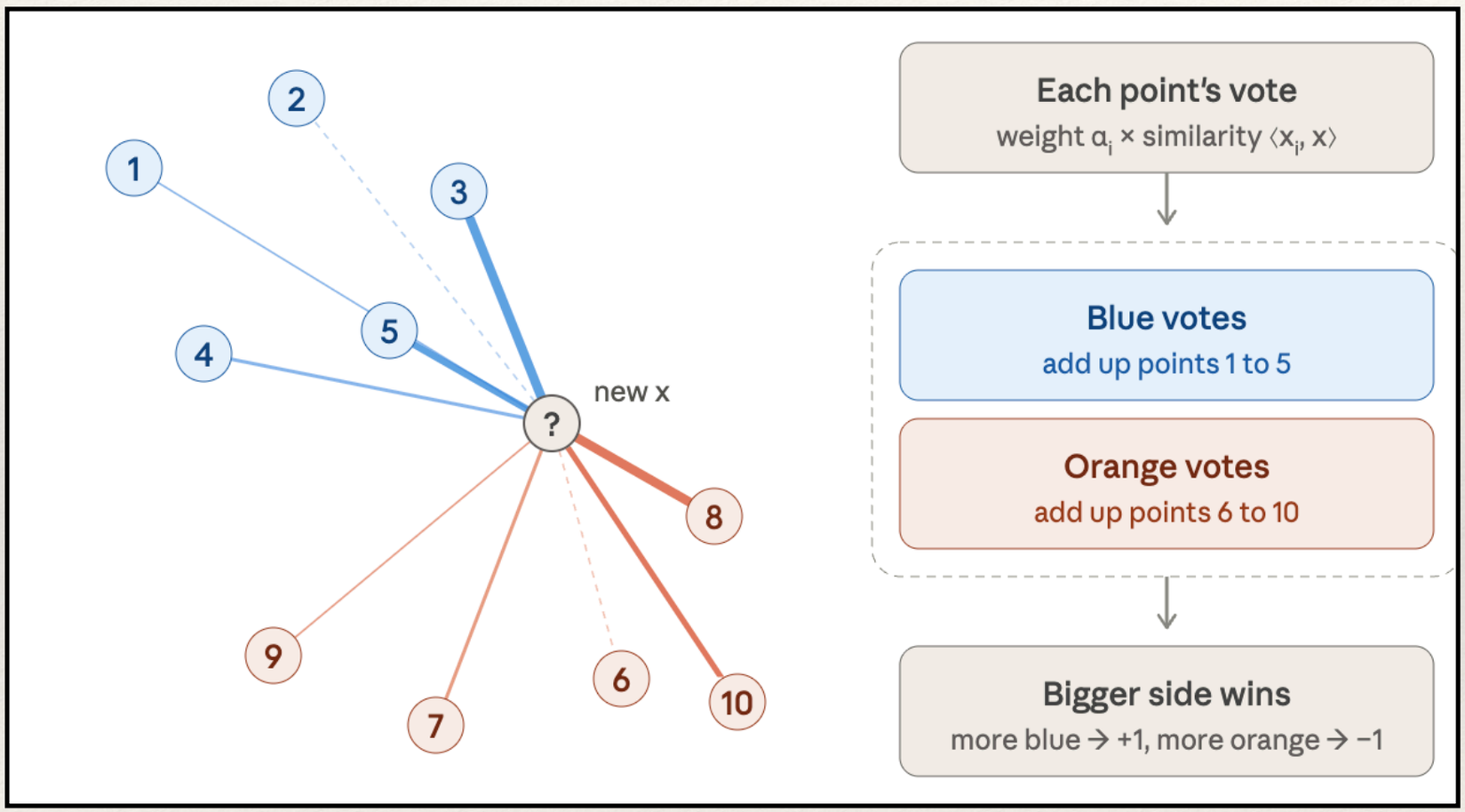
$$w := w + y_j x_j$$

Training point	Times misclassified	Ballots into w
3	3 times	$+ x_3 + x_3 + x_3$
5	2 times	$+ x_5 + x_5$
2	never	nothing
8	3 times	$- x_8 - x_8 - x_8$

$$\mathbf{w} = 3\mathbf{x}_3 + 2\mathbf{x}_5 - 3\mathbf{x}_8 + \cdots = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i$$

$$\hat{y} = \operatorname{sgn}(\mathbf{w} \cdot \mathbf{x}) = \operatorname{sgn}\left(\sum_{i=1}^{n} \alpha_i y_i \langle \mathbf{x}_i, \mathbf{x} \rangle\right)$$

Q1 Analyze: Dual View



α: times misclassified

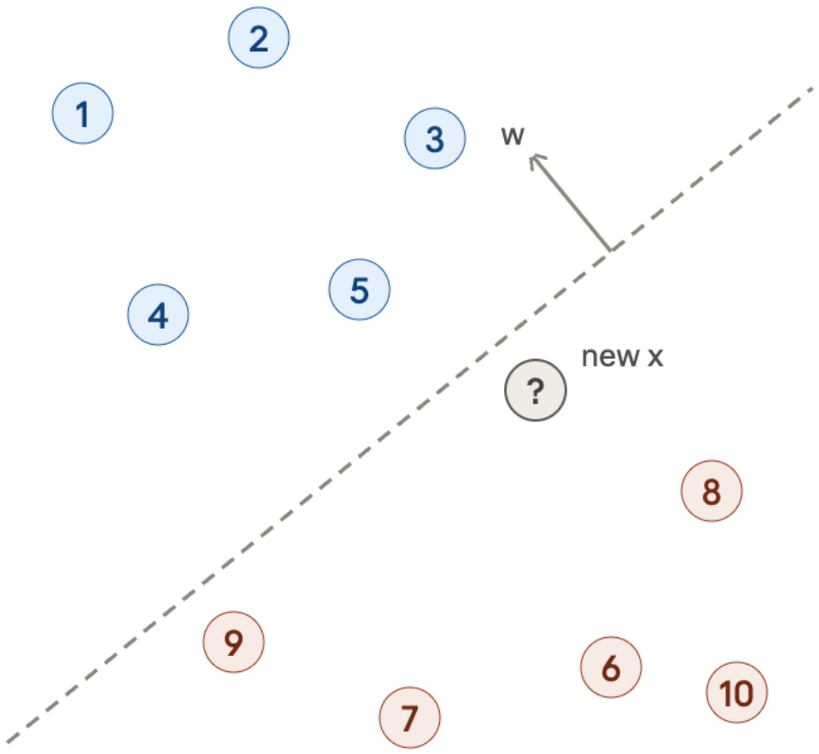
<x_i, x>: similarity

y (label): +1 or -1

$$\hat{y} = \operatorname{sgn}\left(\sum_{i=1}^{n} \alpha_i y_i \langle \mathbf{x}_i, \mathbf{x} \rangle\right)$$

Question 1 (Dual Perceptron)

Review the development of the Perceptron training algorithm from lectures. Now compare this to the algorithm for Perceptron training *in dual form* introduced in the "Kernel Methods" lecture. The two algorithms are very similar but differ in a few crucial places. Provide an explanation of how the dual version of the algorithm relates to the original.



One line, one w
learned from training

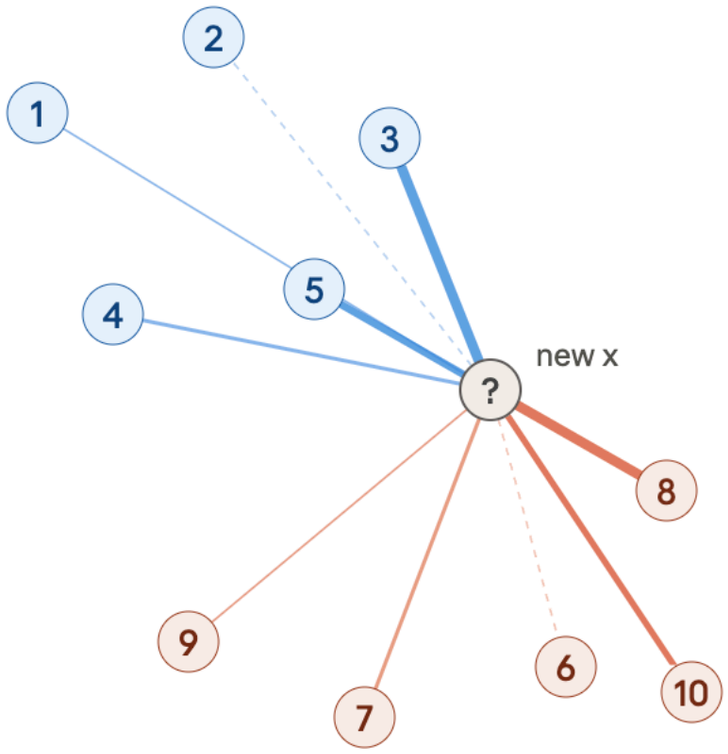
Plug in new x
$s = w_1x_1 + w_2x_2 + w_0$

Which side?
$s > 0 \rightarrow +1, s < 0 \rightarrow -1$

$$\hat{y} = \text{sgn}(\mathbf{w} \cdot \mathbf{x}) = \text{sgn}\left(\sum_{i=1}^{n} \alpha_i y_i \langle \mathbf{x}_i, \mathbf{x} \rangle\right)$$

$$\mathbf{w} = 3\mathbf{x}_3 + 2\mathbf{x}_5 - 3\mathbf{x}_8 + \cdots = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i$$

$$\hat{y} = \text{sgn}\left(\sum_{i=1}^{n} \alpha_i y_i \langle \mathbf{x}_i, \mathbf{x} \rangle\right)$$



Each point's vote
weight α_i × similarity $\langle x_i, x \rangle$

Blue votes
add up points 1 to 5

Orange votes
add up points 6 to 10

Bigger side wins
more blue → +1, more orange → −1

Q1 Answer

View	Original	Dual
Store	W	α

$$\mathbf{w} = 3\mathbf{x}_3 + 2\mathbf{x}_5 - 3\mathbf{x}_8 + \cdots = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i$$

Question 2

Question 2 (Feature Transformations)

Recall that the XOR function (graphed below) is not linearly separable. Show how we can learn the XOR function using a linear classifier after applying a feature transformation to the original dataset. As a concrete example, show how to extend the Dual Perceptron from the previous question to learn the XOR function.

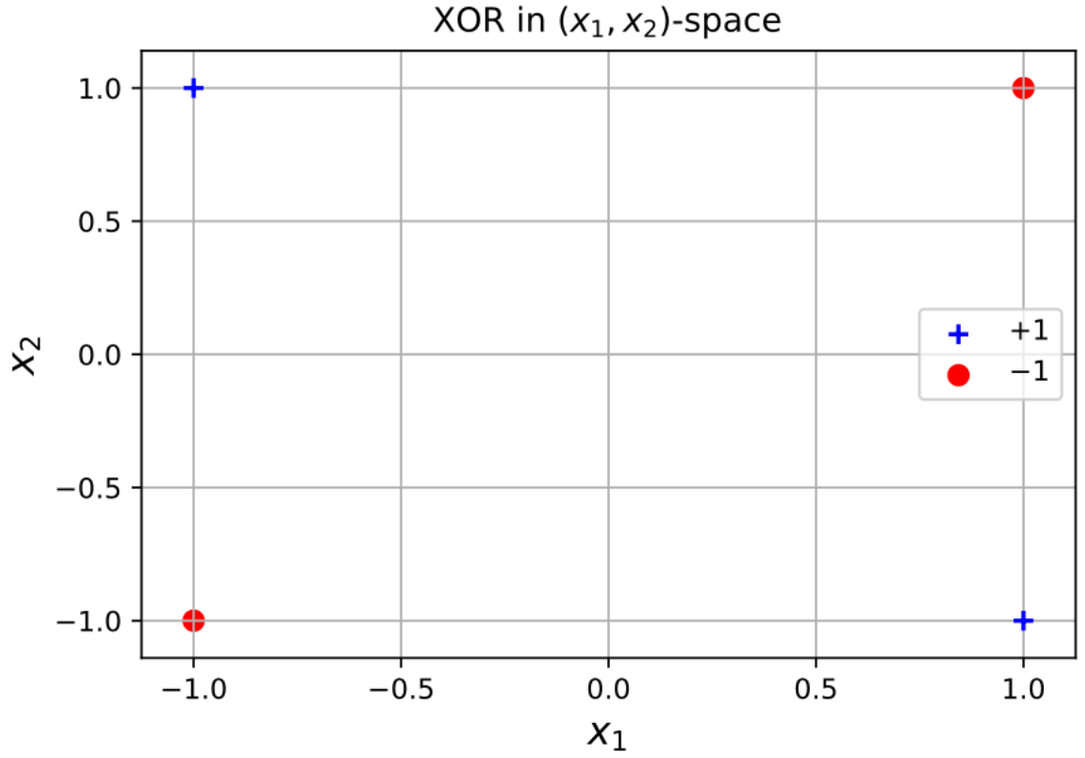


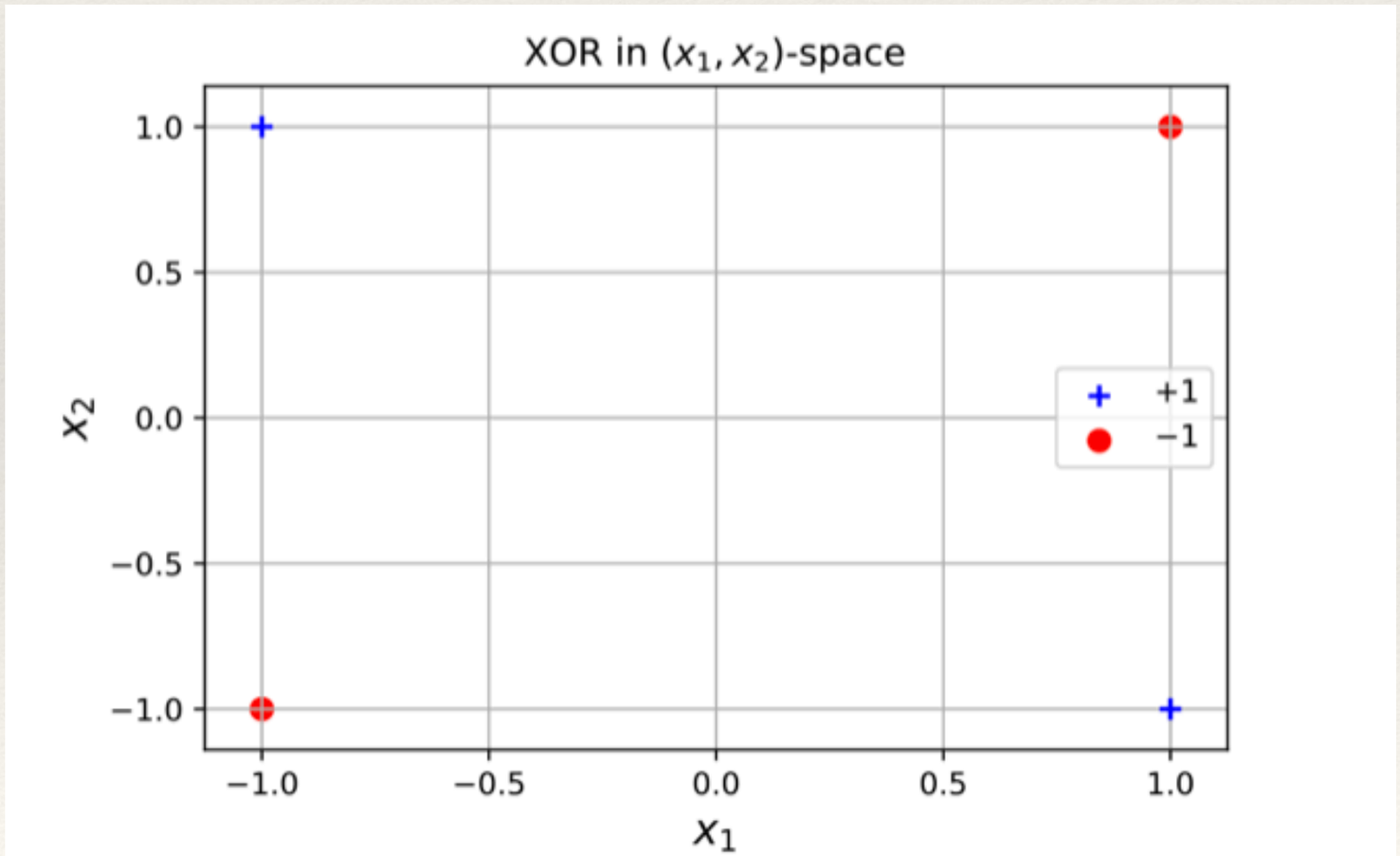
Figure 1: XOR function

Recap: XOR

Both True (1)==> False (-1)

Both False (-1)==> False (-1)

One True (1), One False (1) ==> True (1)



Recap: Feature Transformation

Feature Map / Feature Transformation

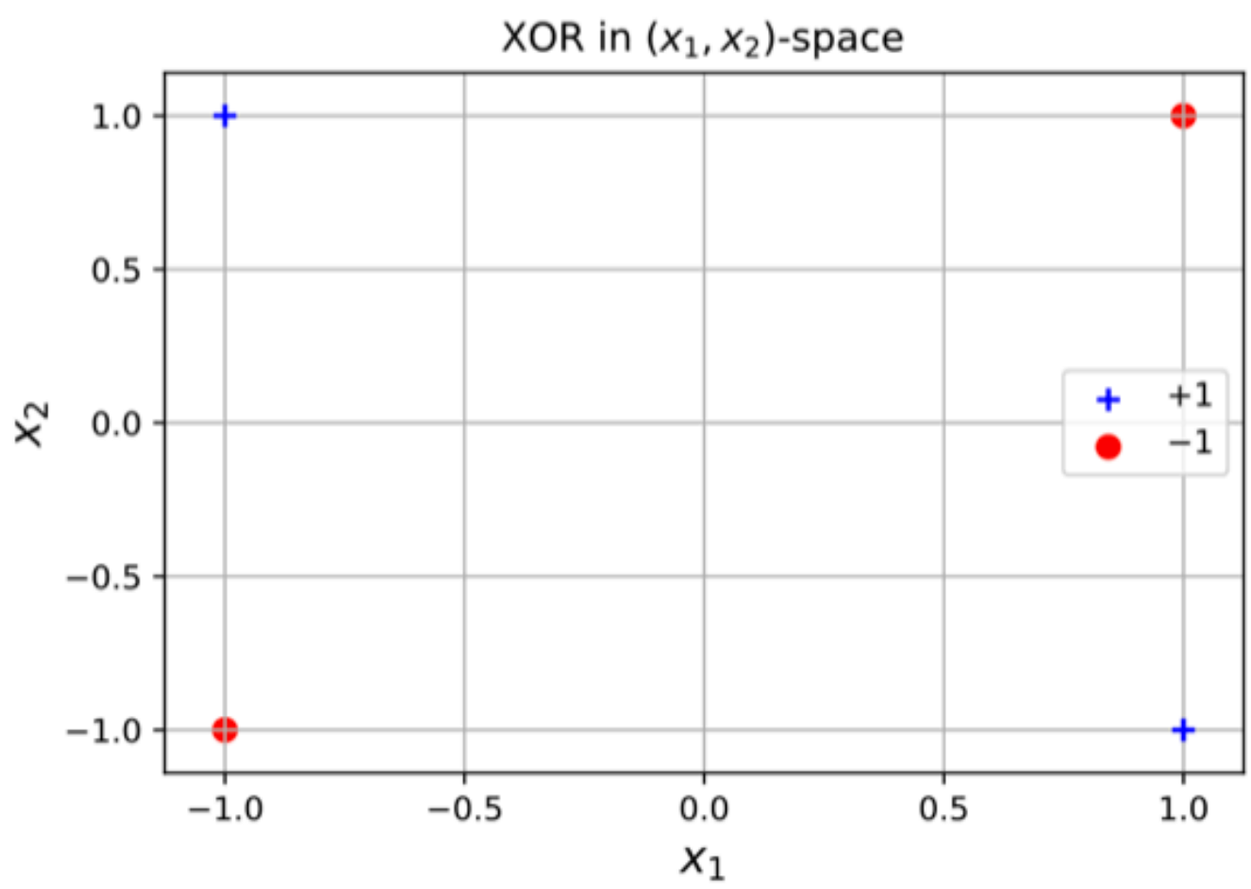
a feature map is any function $\phi : \mathbb{R}^p \rightarrow \mathbb{R}^K$

p=3 $x = [x_1, x_2, x_3]^T,$

$$\phi : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \qquad \phi(x) = \left[\frac{1}{2}(x_1 + x_2), x_3^2\right]^T.$$

Num. of Bedrooms	Num. of Bathrooms	Age of House (years)
4	2	27
1	2	42
4	2	28
4	1	16
4	1	15
3	2	47

Q2 Analyze



$$\phi(\mathbf{x}) = \phi\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \end{bmatrix}.$$

Both True (1)==> False (-1)

Both False (-1)==> False (-1)

One True (1), One False (1) ==> True (1)

Q2 Analyze

$$\phi(\mathbf{x}) = \phi\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \end{bmatrix}.$$

$$\phi\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2} \\ \sqrt{2} \\ 1 \\ 1 \\ \sqrt{2} \end{bmatrix}, \quad \phi\left(\begin{bmatrix} -1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -\sqrt{2} \\ -\sqrt{2} \\ 1 \\ 1 \\ \sqrt{2} \end{bmatrix}, \quad \phi\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -\sqrt{2} \\ \sqrt{2} \\ 1 \\ 1 \\ -\sqrt{2} \end{bmatrix}, \quad \phi\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2} \\ -\sqrt{2} \\ 1 \\ 1 \\ -\sqrt{2} \end{bmatrix}.$$

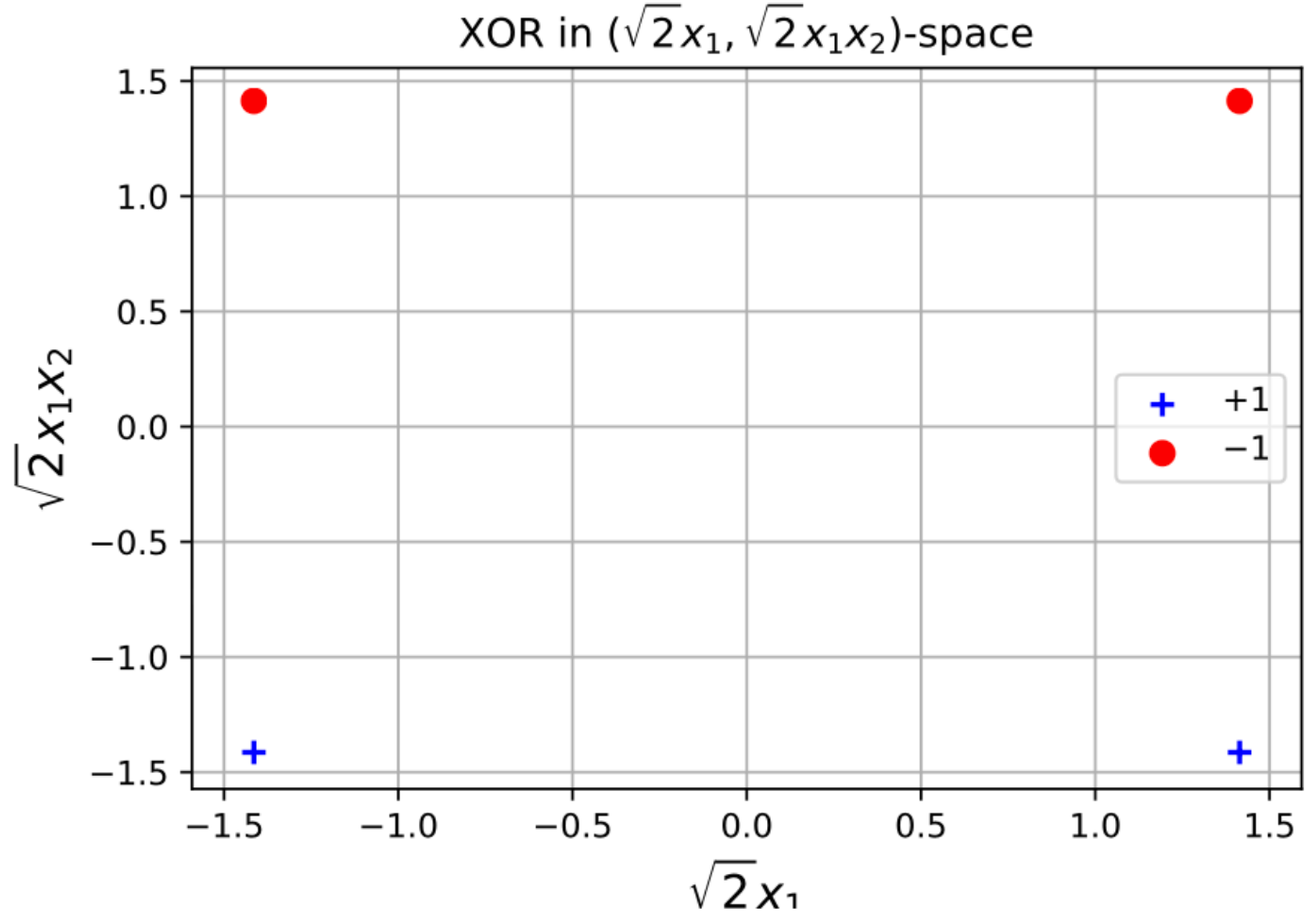
Q2 Visualization

$$\phi(\mathbf{x}) = \phi\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \end{bmatrix}.$$

$$\phi\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2} \\ \sqrt{2} \\ 1 \\ 1 \\ \sqrt{2} \end{bmatrix}, \quad \phi\left(\begin{bmatrix} -1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -\sqrt{2} \\ -\sqrt{2} \\ 1 \\ 1 \\ \sqrt{2} \end{bmatrix}, \quad \phi\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -\sqrt{2} \\ \sqrt{2} \\ 1 \\ 1 \\ -\sqrt{2} \end{bmatrix}, \quad \phi\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2} \\ -\sqrt{2} \\ 1 \\ 1 \\ -\sqrt{2} \end{bmatrix}.$$

Q2 Visualization

Now, we can't visualise 6-dimensional space, but let's pick 2 of the 6 features and plot, namely, we will plot in $(\sqrt{2}x_1, \sqrt{2}x_1x_2)$ (the second and sixth features respectively) space:



After Feature Transformation

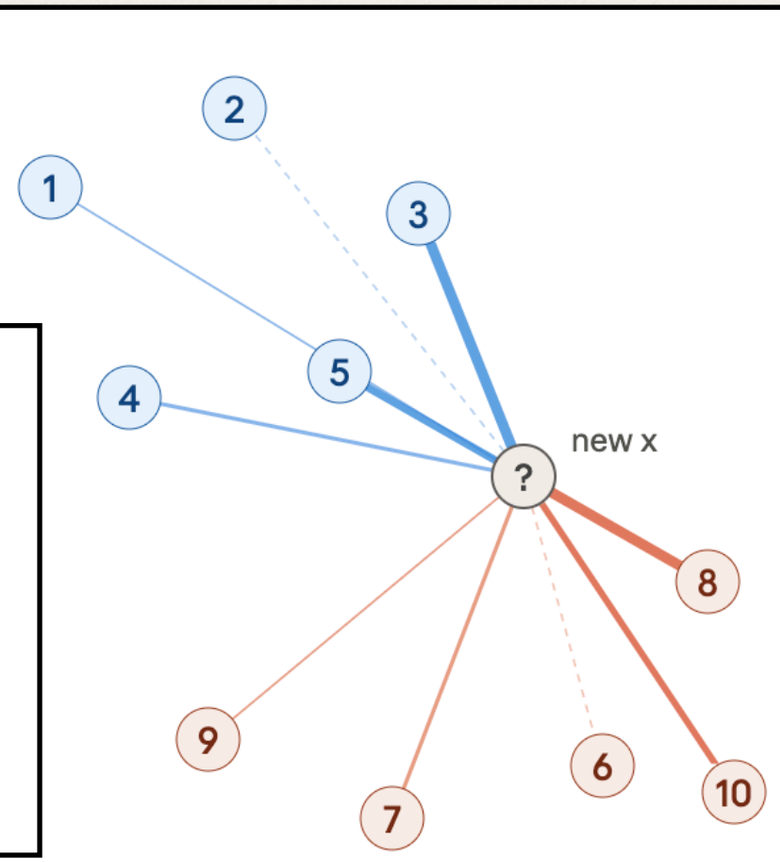
show how to extend the Dual Perceptron

Q2 Analyze: Dual View

Inner Product: <x, x'>

$$\langle \phi(x), \phi(x') \rangle = \quad \ldots$$

$$\phi(\mathbf{x}) = \phi\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \end{bmatrix}.$$



Each point's vote
weight α_i × similarity $\langle x_i, x \rangle$

Blue votes
add up points 1 to 5

Orange votes
add up points 6 to 10

Bigger side wins
more blue → +1, more orange → −1

Q2 Analyze: Dual View

Inner Product: <x, x'>

$$\phi(\mathbf{x}) = \phi\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \end{bmatrix}.$$

$$\begin{aligned} \langle \phi(x), \phi(x') \rangle &= 1 + 2x_1x_1' + 2x_2x_2' + x_1^2{x_1'}^2 + x_2^2{x_2'}^2 + 2x_1x_2x_1'x_2' \\ &= \boxed{(1 + x_1x_1' + x_2x_2')^2} \\ &= \boxed{(1 + \langle x, x' \rangle)^2}. \end{aligned}$$

Question 3

Question 3. (The Kernel Trick)

Using the context of the previous question, discuss the computational issues that arise from having to compute high dimensional feature transformations. Show how these can be mitigated by using the Kernel trick, and use this to extend the dual perceptron learning to kernel perceptron learning.

Q3 Analyze

Computation Issue

Space Comsuming

$$\phi\left(\begin{bmatrix}1\\1\end{bmatrix}\right)=\begin{bmatrix}1\\\sqrt{2}\\\sqrt{2}\\1\\1\\\sqrt{2}\end{bmatrix},\quad \phi\left(\begin{bmatrix}-1\\-1\end{bmatrix}\right)=\begin{bmatrix}1\\-\sqrt{2}\\-\sqrt{2}\\1\\1\\\sqrt{2}\end{bmatrix},\quad \phi\left(\begin{bmatrix}-1\\1\end{bmatrix}\right)=\begin{bmatrix}1\\-\sqrt{2}\\\sqrt{2}\\1\\1\\-\sqrt{2}\end{bmatrix},\quad \phi\left(\begin{bmatrix}1\\-1\end{bmatrix}\right)=\begin{bmatrix}1\\\sqrt{2}\\-\sqrt{2}\\1\\1\\-\sqrt{2}\end{bmatrix}.$$

Time Comsuming

$$\langle x, x' \rangle = x_1 x_1' + x_2 x_2'.$$

$$\langle \phi(x), \phi(x') \rangle = 1 + 2x_1 x_1' + 2x_2 x_2' + x_1^2 {x_1'}^2 + x_2^2 {x_2'}^2 + 2x_1 x_2 x_1' x_2'$$

Q3 Analyze

Computation Issue → Work with Inner Product, not features.

Space Comsuming

$$\phi\left(\begin{bmatrix}1\\1\end{bmatrix}\right)=\begin{bmatrix}1\\\sqrt{2}\\\sqrt{2}\\1\\1\\\sqrt{2}\end{bmatrix},\quad \phi\left(\begin{bmatrix}-1\\-1\end{bmatrix}\right)=\begin{bmatrix}1\\-\sqrt{2}\\-\sqrt{2}\\1\\1\\\sqrt{2}\end{bmatrix},\quad \phi\left(\begin{bmatrix}-1\\1\end{bmatrix}\right)=\begin{bmatrix}1\\-\sqrt{2}\\\sqrt{2}\\1\\1\\-\sqrt{2}\end{bmatrix},\quad \phi\left(\begin{bmatrix}1\\-1\end{bmatrix}\right)=\begin{bmatrix}1\\\sqrt{2}\\-\sqrt{2}\\1\\1\\-\sqrt{2}\end{bmatrix}.$$

Time Comsuming

$$\langle x, x' \rangle = x_1 x_1' + x_2 x_2'.$$

$$\langle \phi(x), \phi(x') \rangle = 1 + 2x_1 x_1' + 2x_2 x_2' + x_1^2 {x_1'}^2 + x_2^2 {x_2'}^2 + 2x_1 x_2 x_1' x_2'$$

Q3 Analyze

Work with Inner Product, not features.

$$\phi(\mathbf{x}) = \phi\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \end{bmatrix}.$$

$$\begin{aligned} \langle \phi(x), \phi(x') \rangle &= 1 + 2x_1x_1' + 2x_2x_2' + x_1^2{x_1'}^2 + x_2^2{x_2'}^2 + 2x_1x_2x_1'x_2' \\ &= \boxed{(1 + x_1x_1' + x_2x_2')^2} \\ &= \boxed{(1 + \langle x, x' \rangle)^2}. \end{aligned}$$

Q3 Analyze

Work with Inner Product, not features.

$$\phi(\mathbf{x}) = \phi\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \end{bmatrix}.$$

$$\begin{aligned}\langle \phi(x), \phi(x') \rangle &= 1 + 2x_1x_1' + 2x_2x_2' + x_1^2x_1'^2 + x_2^2x_2'^2 + 2x_1x_2x_1'x_2' \\ &= \boxed{(1 + x_1x_1' + x_2x_2')^2} \\ &= \boxed{(1 + \langle x, x' \rangle)^2}.\end{aligned}$$

$$\langle x, x' \rangle = x_1x_1' + x_2x_2'. \longrightarrow \boxed{(1 + \langle x, x' \rangle)^2}.$$

Q3 Analyze: Kernel Method

$$\langle x, x' \rangle = x_1 x_1' + x_2 x_2'.$$

$$K(x, x') = (1 + \langle x, x' \rangle)^2$$

$$\hat{y} = \operatorname{sgn}\left(\sum_{i=1}^{n} \alpha_i y_i \langle \mathbf{x}_i, \mathbf{x} \rangle\right)$$

$$\hat{y} = \operatorname{sgn}\left(\sum_{i} \alpha_i y_i K(x_i, x)\right)$$

Dual perceptron

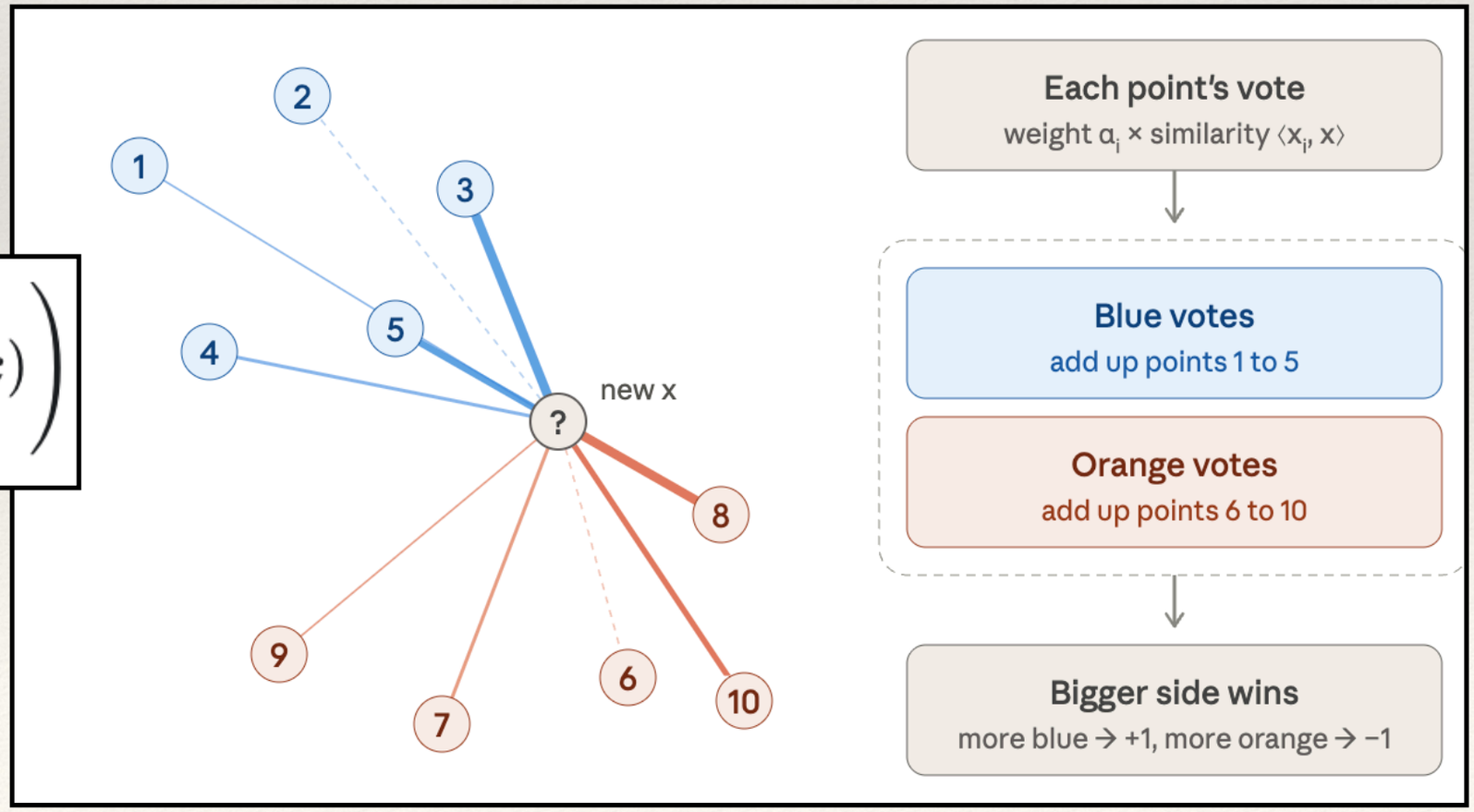
Kernel perceptron

Q3 Intuition: Kernel Method

A kernel method predicts a new point
by measuring its relationship with relevant points
and combining their contributions.

Example: Kernel Perceptron (Similarity-Weighted Voting)

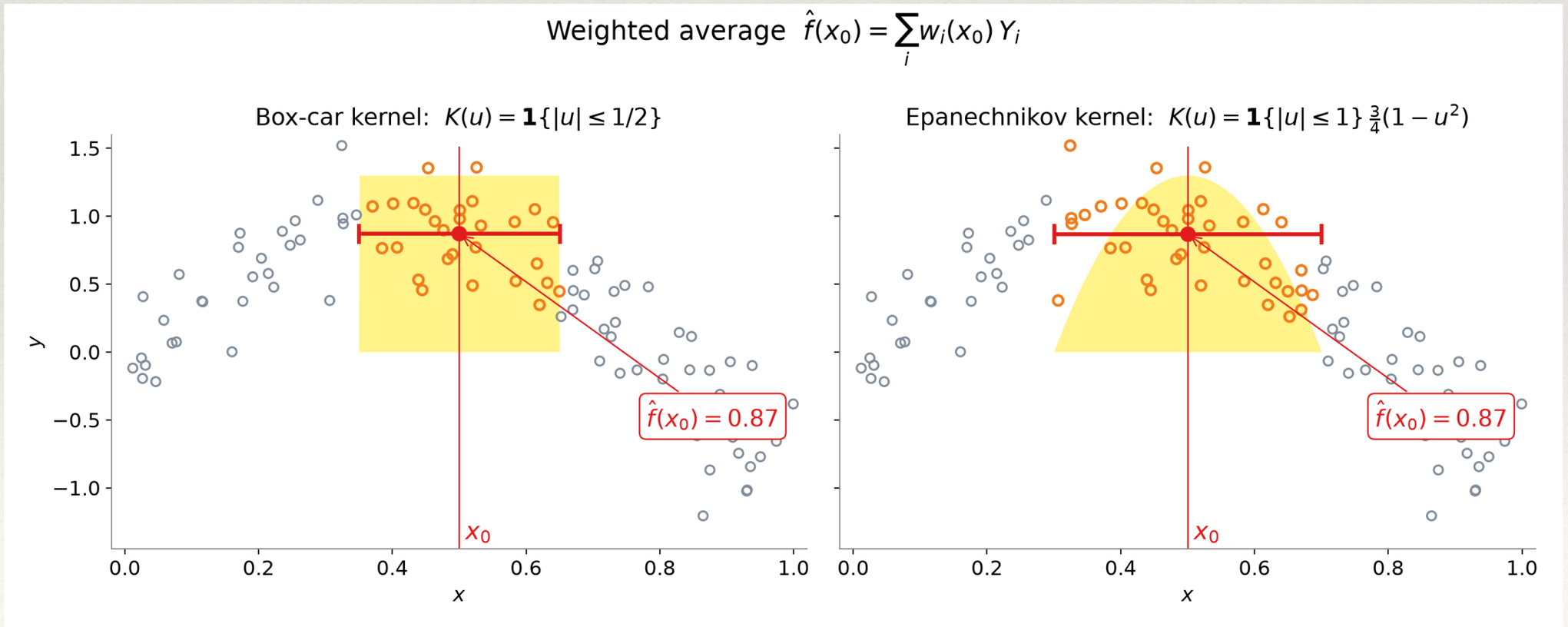
$$\hat{y} = \operatorname{sgn}\left(\sum_i \alpha_i y_i K(x_i, x)\right)$$



Q3 Intuition: Kernel Method

A kernel method predicts a new point
by measuring its relationship with relevant points
and combining their contributions.

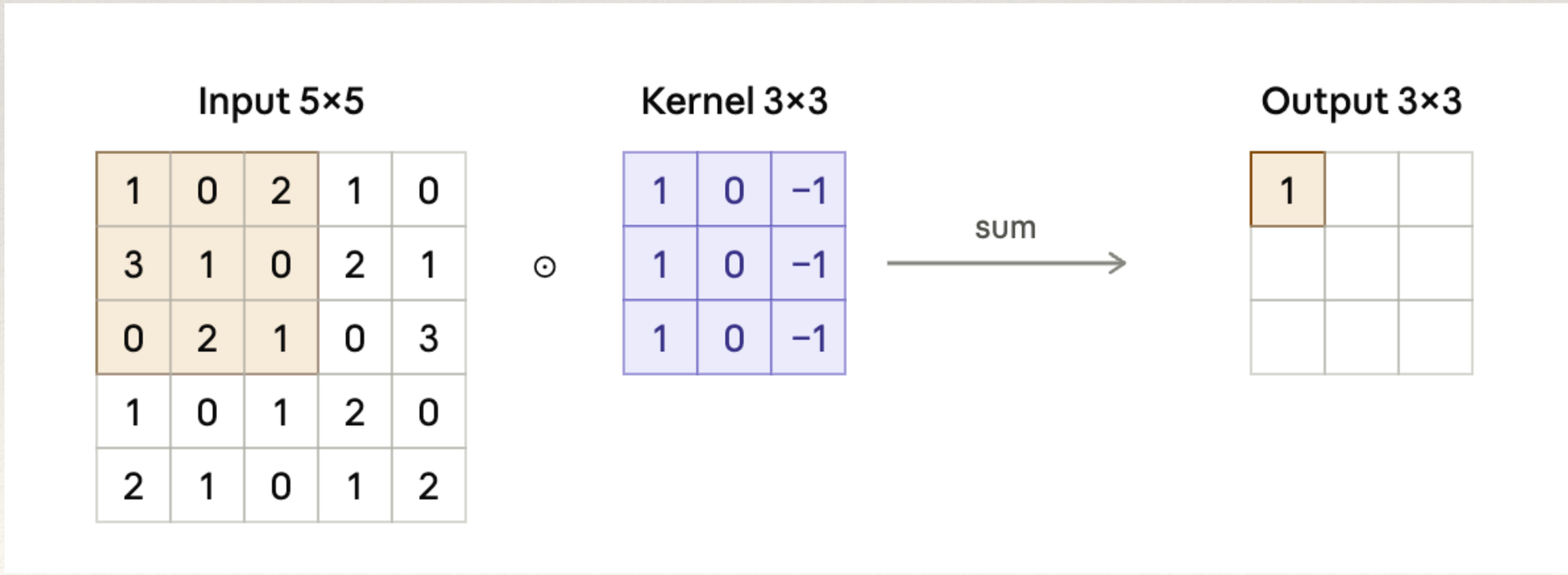
Example (week 5): Kernel Smoothing (Weighted Average)



Q3 Intuition: Kernel Method

A kernel method predicts a new point
by measuring its relationship with relevant points
and combining their contributions.

Example: CNN



Question 4

Question 4. (Kernels and their Feature Representations)

In this question, we will show how the choice of kernel gives us different feature transformations. Note that in practice, we will simply choose a kernel and not be too concerned with the exact feature transformation, but it is important to know that different kernels correspond to different representations.

(a) Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ (i.e. $\mathbf{x}$ and $\mathbf{y}$ are two dimensional vectors), and consider the kernel

$$k(\mathbf{x}, \mathbf{y}) = (2\langle \mathbf{x}, \mathbf{y} \rangle + 3)^3.$$

Compute the feature vector $\phi(\mathbf{x})$ correpsonding to this kernel. (In other words, the feature representation of $\mathbf{x}$ and $\mathbf{y}$ such that $\langle \phi(\mathbf{x}), \phi(\mathbf{y}) \rangle = k(\mathbf{x}, \mathbf{y})$)

Kernel Function ⟺ Feature Transformation

Recap Q3

$$\phi(\mathbf{x}) = \phi\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \end{bmatrix}$$

Feature Transformation

$$\langle \phi(x), \phi(x') \rangle = 1 + 2x_1x_1' + 2x_2x_2' + x_1^2x_1'^2 + x_2^2x_2'^2 + 2x_1x_2x_1'x_2'$$

$$= \boxed{(1 + x_1x_1' + x_2x_2')^2}$$

$$= \boxed{(1 + \langle x, x' \rangle)^2}.$$

$$\langle x, x' \rangle = x_1x_1' + x_2x_2'. \longrightarrow$$

Kernel Function

$$\boxed{(1 + \langle x, x' \rangle)^2}.$$

Q4a Analyze

(a) Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ (i.e. $\mathbf{x}$ and $\mathbf{y}$ are two dimensional vectors), and consider the kernel

$$k(\mathbf{x}, \mathbf{y}) = (2\langle \mathbf{x}, \mathbf{y} \rangle + 3)^3.$$

Compute the feature vector $\phi(\mathbf{x})$ correpsonding to this kernel. (In other words, the feature representation of $\mathbf{x}$ and $\mathbf{y}$ such that $\langle \phi(\mathbf{x}), \phi(\mathbf{y}) \rangle = k(\mathbf{x}, \mathbf{y})$)

Step 1: Expand the kernel

$$\begin{aligned}
k(x, y) &= (2\langle x, y \rangle + 3)^3 \\
&= \big(2(x_1 y_1 + x_2 y_2) + 3\big)^3 \\
&= 8(x_1 y_1 + x_2 y_2)^3 + 36(x_1 y_1 + x_2 y_2)^2 + 54(x_1 y_1 + x_2 y_2) + 27 \\
&= 8\big((x_1 y_1)^3 + 3(x_1 y_1)^2 (x_2 y_2) + 3(x_1 y_1)(x_2 y_2)^2 + (x_2 y_2)^3\big) \\
&\quad + 36\big((x_1 y_1)^2 + 2 x_1 y_1 x_2 y_2 + (x_2 y_2)^2\big) \\
&\quad + 54 x_1 y_1 + 54 x_2 y_2 + 27 \\
&= 8 x_1^3 y_1^3 + 8 x_2^3 y_2^3 + 24 x_1^2 y_1^2 x_2 y_2 + 24 x_1 y_1 x_2^2 y_2^2 \\
&\quad + 36 x_1^2 y_1^2 + 36 x_2^2 y_2^2 + 72 x_1 y_1 x_2 y_2 + 54 x_1 y_1 + 54 x_2 y_2 + 27.
\end{aligned}$$

Q4a Analyze

(a) Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$ (i.e. $\mathbf{x}$ and $\mathbf{y}$ are two dimensional vectors), and consider the kernel

$$k(\mathbf{x}, \mathbf{y}) = (2\langle \mathbf{x}, \mathbf{y} \rangle + 3)^3.$$

Compute the feature vector $\phi(\mathbf{x})$ correpsonding to this kernel. (In other words, the feature representation of $\mathbf{x}$ and $\mathbf{y}$ such that $\langle \phi(\mathbf{x}), \phi(\mathbf{y}) \rangle = k(\mathbf{x}, \mathbf{y})$)

Step 2: Factor each term into an x-part and a y-part.

$$8x_1^3y_1^3 + 8x_2^3y_2^3 + 24x_1^2y_1^2x_2y_2 + 24x_1y_1x_2^2y_2^2$$
$$+ 36x_1^2y_1^2 + 36x_2^2y_2^2 + 72x_1y_1x_2y_2 + 54x_1y_1 + 54x_2y_2 + 27.$$

Q4a Answer

$$8x_1^3y_1^3 + 8x_2^3y_2^3 + 24x_1^2y_1^2x_2y_2 + 24x_1y_1x_2^2y_2^2 + 36x_1^2y_1^2 + 36x_2^2y_2^2 + 72x_1y_1x_2y_2 + 54x_1y_1 + 54x_2y_2 + 27.$$

$$\phi^T(x) = [\sqrt{8}x_1^3, \sqrt{8}x_2^3, \sqrt{24}x_1^2x_2, \sqrt{24}x_1x_2^2, \sqrt{36}x_1^2, \sqrt{36}x_2^2, \sqrt{72}x_1x_2, \sqrt{54}x_1, \sqrt{54}x_2, \sqrt{27}]$$

Question 5

Question 5 (More of the Kernel Trick)

You are told that the "kernel trick" means that a non-linear mapping can be realised from the original data representation to a new, implicit feature space simply by defining a kernel function on dot products of pairs of instances from the original data. To see why this is so, you take two instances $\mathbf{x} = [1, 2]^T$ and $\mathbf{y} = [3, 2]^T$, and take their dot product $\langle \mathbf{x}, \mathbf{y} \rangle$ and obtain the answer 7. Clearly, raising this dot product to the power of two will give $(\langle \mathbf{x}, \mathbf{y} \rangle)^2 = 49$. Now expand out this expression to show that this is the same answer you would have obtained if you had simply done a set of feature transformations on the original data.

2D feature vectors: $\mathbf{x} = [1, 2]^T$ $\mathbf{y} = [3, 2]^T$

Kernel Function ⟺ Feature Transformation

$(\langle \mathbf{x}, \mathbf{y} \rangle)^2$ …

Q5 Analyze

$$(\langle \mathbf{x}, \mathbf{y} \rangle)^2$$

Step 1: Expand the kernel

$$\big(\langle x, y \rangle\big)^2 = (x_1 y_1 + x_2 y_2)^2.$$

Step 2: Factor each term into an x-part and a y-part.

Q5 Analyze

$$(\langle \mathbf{x}, \mathbf{y} \rangle)^2$$

Step 1: Expand the kernel

$$(\langle x, y \rangle)^2 = (x_1 y_1 + x_2 y_2)^2.$$

$$= x_1^2 y_1^2 + x_2^2 y_2^2 + 2 x_1 y_1 x_2 y_2$$

Step 2: Factor each term into an x-part and a y-part.

$$\begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2} x_1 x_2 \end{bmatrix}, \begin{bmatrix} y_1^2 \\ y_2^2 \\ \sqrt{2} y_1 y_2 \end{bmatrix}$$

Q5 Analyze: Kernal Method

2D feature vectors: $\mathbf{x} = [1, 2]^T$ $\mathbf{y} = [3, 2]^T$

Kernel Function

$$\langle x, y \rangle = 1 \cdot 3 + 2 \cdot 2 = 7, \qquad \left(\langle x, y \rangle\right)^2 = 7^2 = \boxed{49}.$$

Q5 Analyze: Feature Transformation

2D feature vectors: $\mathbf{x} = [1, 2]^T$ $\mathbf{y} = [3, 2]^T$

$$\phi(x) = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2}x_1x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 2\sqrt{2} \end{bmatrix}, \qquad \phi(y) = \begin{bmatrix} y_1^2 \\ y_2^2 \\ \sqrt{2}y_1y_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \\ 6\sqrt{2} \end{bmatrix}.$$

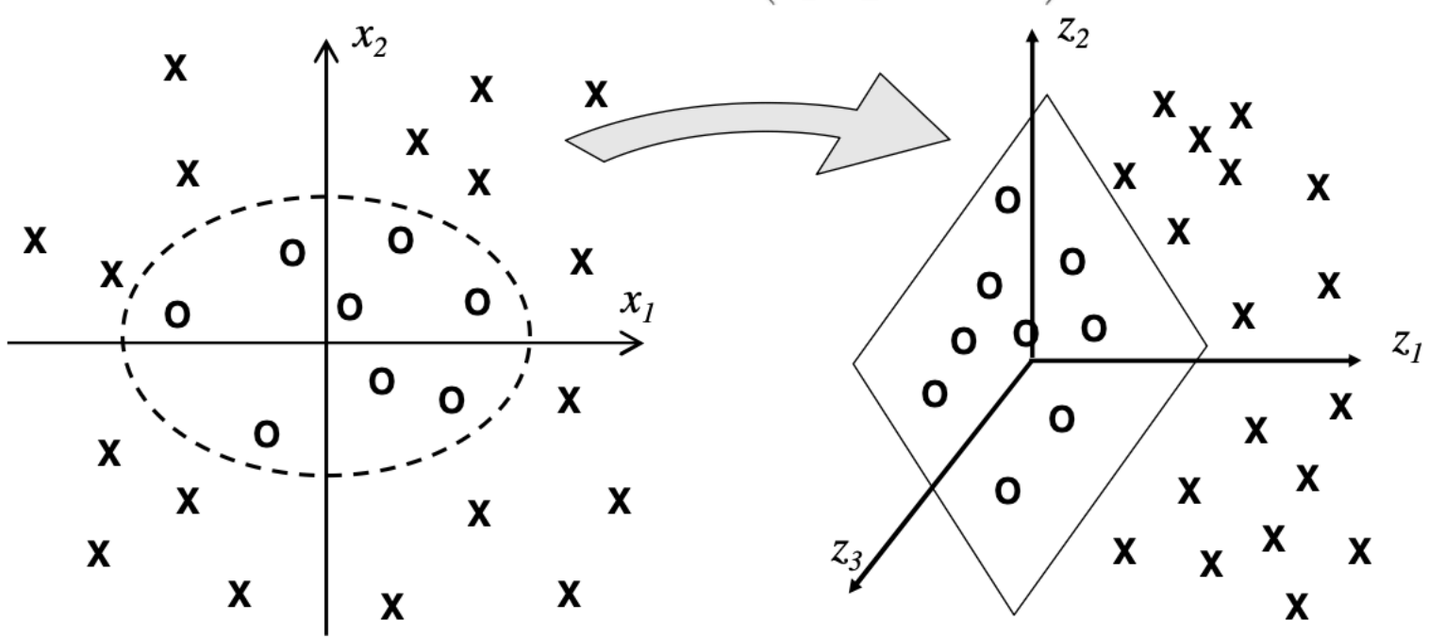
$$\langle \phi(x), \phi(y) \rangle = 1 \cdot 9 + 4 \cdot 4 + (2\sqrt{2})(6\sqrt{2}) = 9 + 16 + 24 = \boxed{49}.$$

Question 6

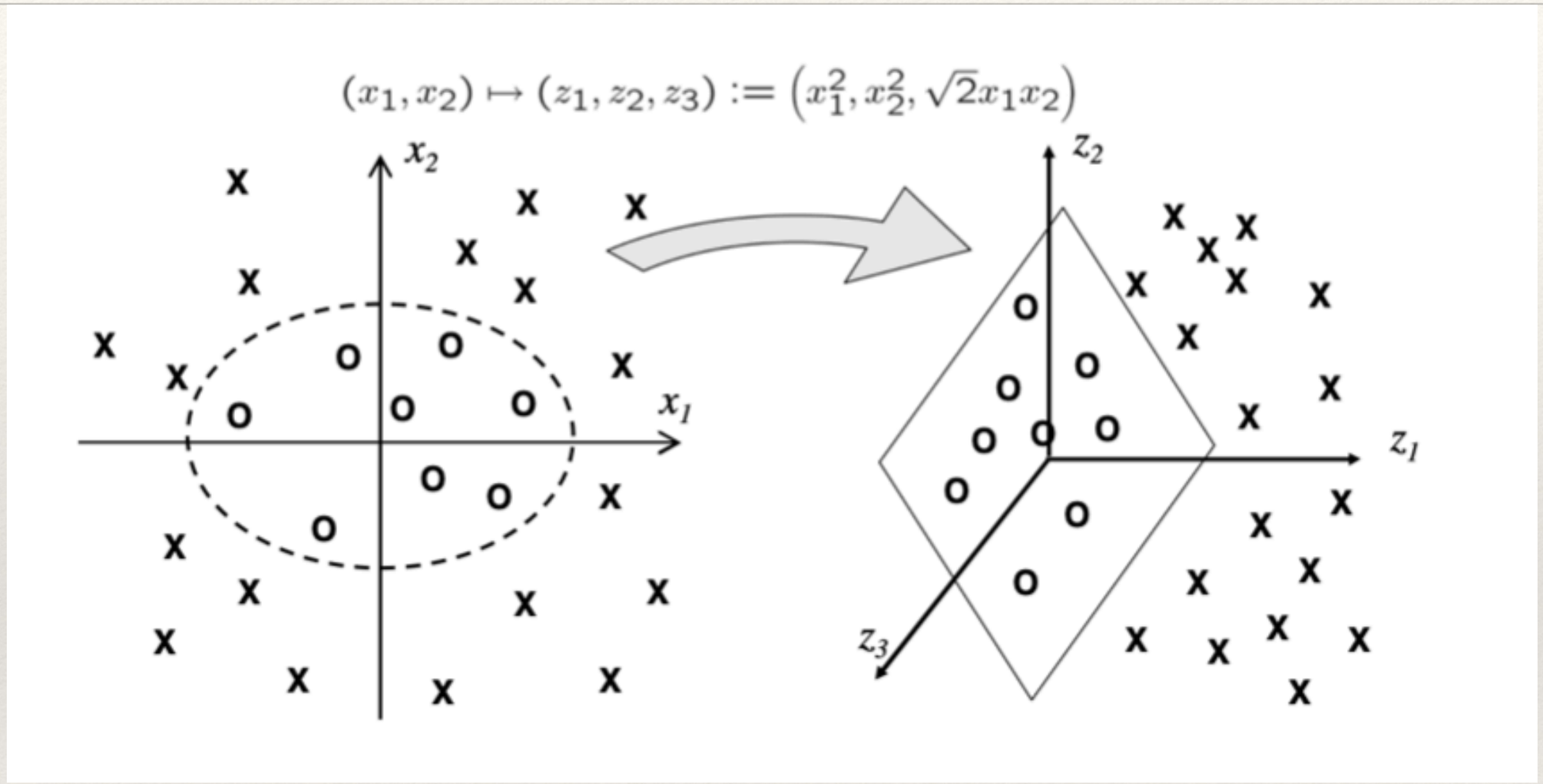
Question 6 (More Feature Transformations)

Consider the following depiction of a feature transformation from two dimensional space ($\mathbb{R}^2$) to three dimensional space ($\mathbb{R}^3$). Why would we use such a transformation? Generate one example from the $\circ$ class in the original space, and another example from the $\times$ class in the original space, and show their transformed values in the new space.

$$(x_1, x_2) \mapsto (z_1, z_2, z_3) := \left(x_1^2, x_2^2, \sqrt{2}x_1x_2\right)$$



Q6 Analyze



We need a Circle:

$$x_1^2 + x_2^2 = c.$$

Q6 Analyze

We need a Circle: $x_1^2 + x_2^2 = c.$ (Not Linear)

Feature Transformation: $(x_1, x_2) \mapsto (z_1, z_2, z_3) := \left(x_1^2, x_2^2, \sqrt{2}x_1x_2\right)$

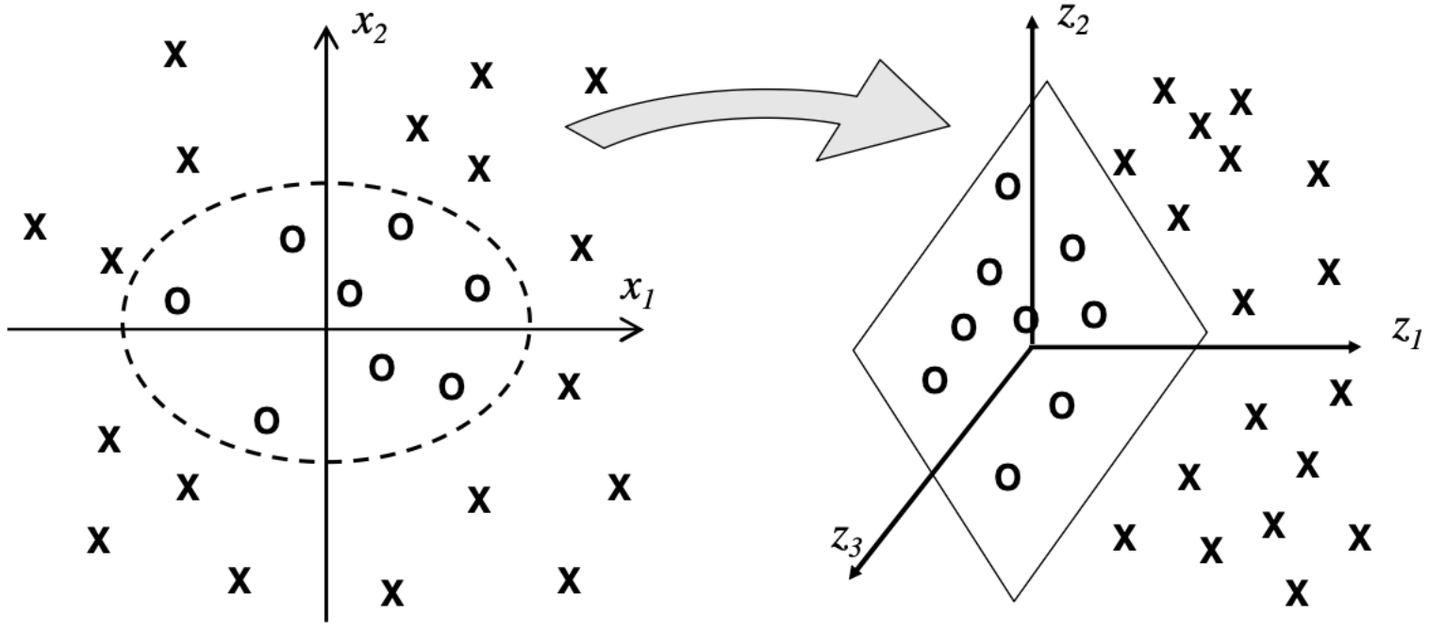
Circle —> Line $z_1 + z_2 = c$ (Linear)

Q6 Kernel Function

Question 6 (More Feature Transformations)

Consider the following depiction of a feature transformation from two dimensional space ($\mathbb{R}^2$) to three dimensional space ($\mathbb{R}^3$). Why would we use such a transformation? Generate one example from the $\circ$ class in the original space, and another example from the $\times$ class in the original space, and show their transformed values in the new space.

$$(x_1, x_2) \mapsto (z_1, z_2, z_3) := \left(x_1^2, x_2^2, \sqrt{2}x_1x_2\right)$$



Kernel Function ⟺ Feature Transformation

$$\left(x_1^2, x_2^2, \sqrt{2}x_1x_2\right)$$

Q6 Analyze: Calculate Kernel Function

Kernel Function  Feature Transformation

$$\left(x_1^2, x_2^2, \sqrt{2}x_1x_2\right)$$

Step 1: Compute the inner product

Step 2: Simplify the result

Q6 Calculation

Kernel Function ⟺ Feature Transformation

$$\left(x_1^2, x_2^2, \sqrt{2}x_1x_2\right)$$

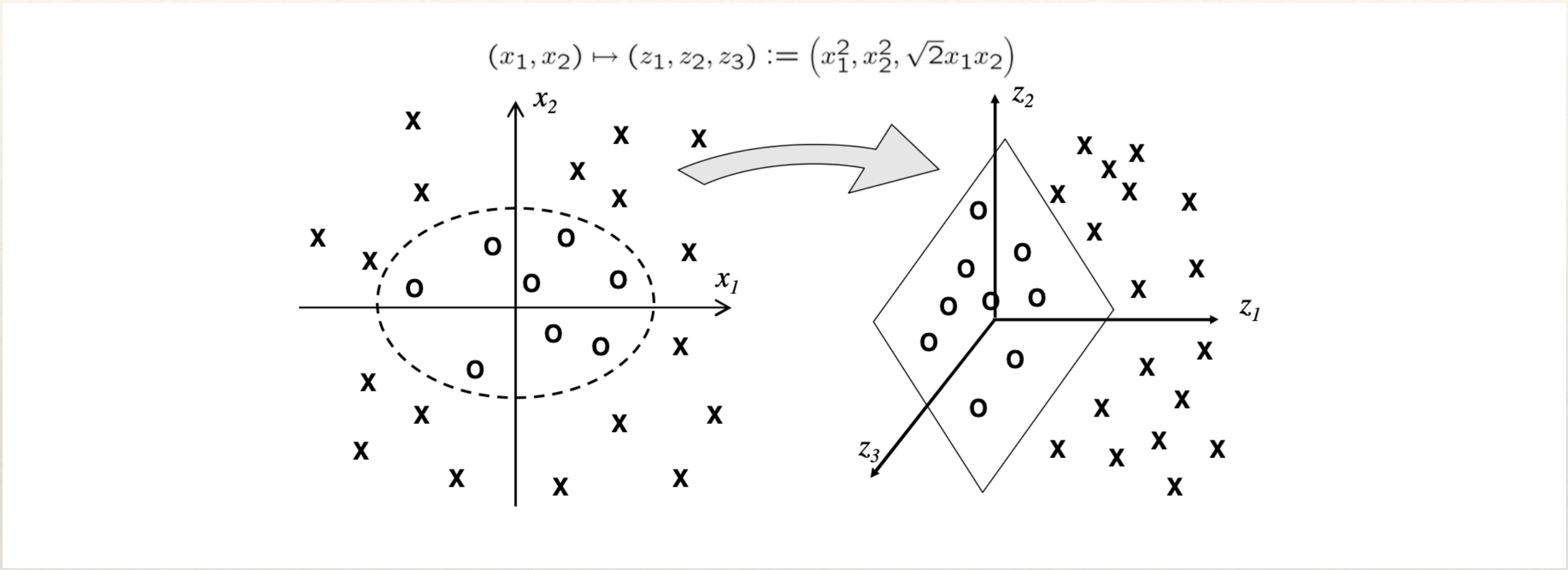
Step 1: Compute the inner product

Step 2: Simplify the result

$$\begin{aligned} K(x,y) &= \langle \phi(x), \phi(y) \rangle \\ &= x_1^2 y_1^2 + x_2^2 y_2^2 + (\sqrt{2}x_1x_2)(\sqrt{2}y_1y_2) \\ &= x_1^2 y_1^2 + 2x_1y_1x_2y_2 + x_2^2 y_2^2 \\ &= (x_1y_1 + x_2y_2)^2 \\ &= \langle x, y \rangle^2. \end{aligned}$$

$$K(x,y) = \|x\|^2 \|y\|^2 \cos^2\theta.$$

Q6 Intuition



$$K(x, y) = \|x\|^2 \|y\|^2 \cos^2 \theta.$$

Thank you!